

SOC INTERNSHIP REPORT

Week 1 – Task 3 (SOC)

Virtual Security Lab Setup & Windows Fundamentals

Submitted By: MANOJ

Role: Cyber Security Intern

Internship Organisation: TechBiz Security

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1. Introduction

This report documents the completion of Week 1, Task 3 of the SOC internship. The task focuses on validating a practical virtual cybersecurity lab environment and completing the TryHackMe “Windows Fundamentals 1” room. A controlled virtual lab is an important resource for a Security Operations Centre (SOC) analyst, as it provides an isolated environment where systems can be examined, network communication can be tested, and security-related activities such as monitoring, log analysis, and investigation can be practised without risk to production systems.

2. Task Objective

The objective of this task was to verify that the existing Kali Linux and Windows Server 2022 virtual machines were operational, confirm network communication between the two systems, and complete the required TryHackMe Windows Fundamentals 1 room.

3. Existing Lab Environment

The Kali Linux and Windows Server 2022 virtual machines used for this task were already installed and configured prior to the start of this task. The internship administrator confirmed that reinstallation of the virtual machines was not required. Accordingly, this task focused on the practical validation and use of the existing lab environment rather than on new installation work. The following components make up the lab environment:

- Oracle VirtualBox is used as the virtualization platform hosting both virtual machines.
- Kali Linux is used as the Linux-based security analysis system.
- Windows Server 2022 is used as the Windows-based analysis/target system.
- No virtual machines were reinstalled during this task.

4. Lab Environment Overview

Table 1 below summarises the components of the lab environment and their purpose.

Component	Purpose
Oracle VirtualBox	Virtualization platform
Kali Linux	Security testing and analysis system
Windows Server 2022	Windows-based analysis/target system
Virtual Network	Enables communication between the lab systems

5. Kali Linux VM Verification

The Kali Linux virtual machine was confirmed to be running successfully inside Oracle VirtualBox. The `ip a` command was used within the terminal to inspect the network interface configuration. The active Ethernet interface (eth0) displayed an IPv4 address of 10.206.175.48. A ping test was then performed from Kali Linux to the Windows Server 2022 system at 10.206.175.18. The test received replies for all packets sent, with 0% packet loss, confirming that the Kali Linux VM had a working network connection to the Windows Server 2022 VM. Supporting evidence for this verification is provided in Figure 1 (Section 13).

6. Windows Server 2022 VM Verification

The Windows Server 2022 virtual machine was confirmed to be running successfully inside Oracle VirtualBox. The ipconfig command was used in the Administrator Command Prompt to inspect the network configuration. The Ethernet adapter displayed an IPv4 address of 10.206.175.18 with a subnet mask of 255.255.255.0. A ping test was then performed from Windows Server 2022 to the Kali Linux system at 10.206.175.48. The test showed 4 packets sent, 4 received, and 0% packet loss, confirming a working network connection from the Windows Server 2022 VM to the Kali Linux VM. Supporting evidence for this verification is provided in Figure 2 (Section 13).

7. IP Configuration and Network Connectivity Testing

Bidirectional ICMP connectivity between the two virtual machines was validated in both directions: from Kali Linux (10.206.175.48) to Windows Server 2022 (10.206.175.18), and from Windows Server 2022 (10.206.175.18) to Kali Linux (10.206.175.48). Both tests completed successfully with no packet loss, confirming that the two systems can communicate reliably over the configured virtual network.

8. Connectivity Test Results

Table 2 summarises the connectivity test results between the two virtual machines.

Source	Destination	Method	Result
Kali Linux (10.206.175.48)	Windows Server 2022 (10.206.175.18)	ping	Successful, 0% packet loss
Windows Server 2022 (10.206.175.18)	Kali Linux (10.206.175.48)	ping	Successful, 0% packet loss

The successful bidirectional communication confirms that the Kali Linux and Windows Server 2022 virtual machines can communicate over the configured virtual network, which is a prerequisite for future SOC-related activities involving these systems.

9. TryHackMe Windows Fundamentals 1

As part of this task, the required TryHackMe “Windows Fundamentals 1” room was completed. This room provides foundational knowledge of the Windows environment, covering topics such as Windows editions, the desktop/GUI, general Windows basics, the file system, and the Windows System32 folder structure, as reflected in the completed task list. Supporting evidence of completion is provided in Figures 3 and 4 (Section 13).

10. Practical Work Performed

- Accessed the existing Kali Linux virtual machine.
- Verified its network configuration using the ip a command.
- Identified the IPv4 address 10.206.175.48.
- Tested connectivity from Kali Linux to Windows Server 2022.
- Accessed the existing Windows Server 2022 virtual machine.
- Verified its network configuration using the ipconfig command.
- Identified the IPv4 address 10.206.175.18.

- Tested connectivity from Windows Server 2022 to Kali Linux.
- Confirmed 0% packet loss in both directions.
- Completed the TryHackMe Windows Fundamentals 1 room.
- Captured screenshots as evidence of the practical work performed.

11. SOC Relevance of the Lab

This lab environment supports future SOC activities in the following ways:

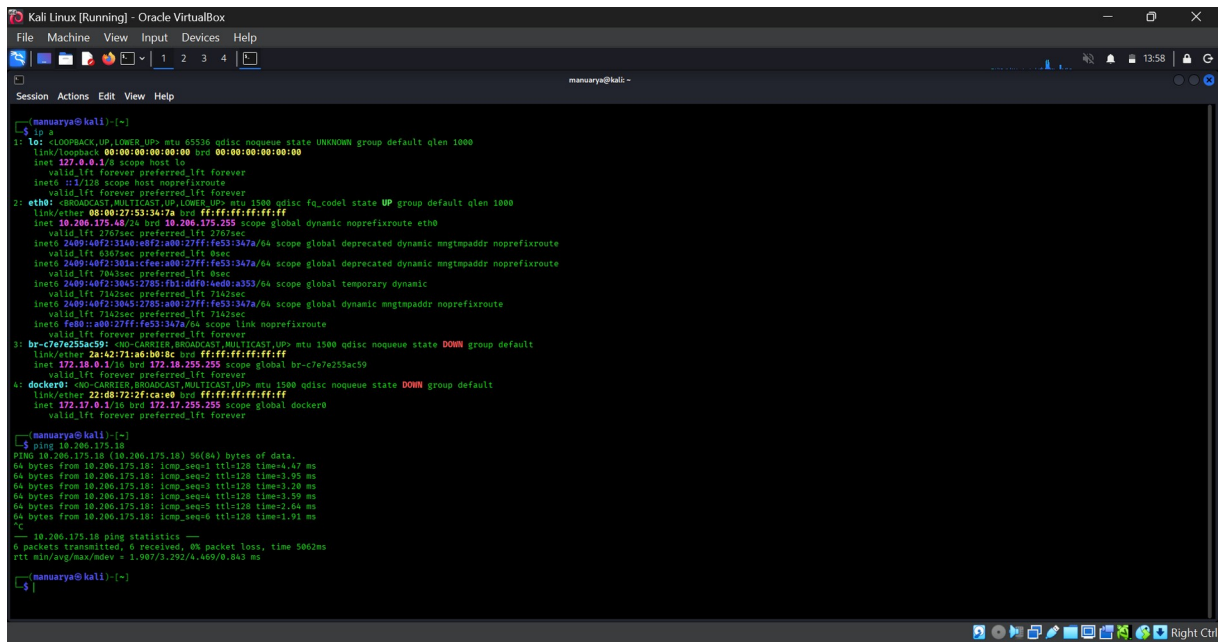
- Understanding Windows systems and their file structure.
- Investigating Windows security events and logs.
- Practising endpoint monitoring.
- Generating and analysing security-related activity.
- Connecting security tools such as SIEM and endpoint monitoring agents in future tasks.
- Performing controlled investigation activities between an analysis machine and a Windows endpoint.

12. Skills and Knowledge Gained

- Virtual machine environment validation.
- Basic Linux network configuration inspection.
- Basic Windows network configuration inspection.
- IPv4 addressing.
- ICMP/ping connectivity testing.
- Windows fundamentals.
- Practical cybersecurity lab preparation.
- Understanding of how a controlled lab can support SOC investigations.

13. Evidence and Screenshot Documentation

The following screenshots provide direct evidence of the practical work performed during this task.



```

manuara@kali:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:53:34:7a brd ff:ff:ff:ff:ff:ff
    inet 10.206.175.48/24 brd 10.206.175.255 scope global dynamic noprefixroute eth0
        valid_lft 2705sec preferred_lft 2705sec
    inet6 2409:40f2:3045:e8f2:a00:27ff:fe53:347a/64 scope global deprecated dynamic mngtmpaddr noprefixroute
        valid_lft 6367sec preferred_lft 6367sec
    inet6 2409:40f2:3045:cfce:a00:27ff:fe53:347a/64 scope global deprecated dynamic mngtmpaddr noprefixroute
        valid_lft 7043sec preferred_lft 7043sec
    inet6 2409:40f2:3045:2785:a00:27ff:fe53:347a/64 scope global temporary dynamic
        valid_lft 7345sec preferred_lft 7345sec
    inet6 2409:40f2:3045:2785:a00:27ff:fe53:347a/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 7345sec preferred_lft 7345sec
    inet6 fe80::a00:27ff:fe53:347a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: br-c7e255ac59: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 2a:42:71:a6:b0:8c brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-c7e255ac59
        valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 72:42:72:2f:ca:00 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever

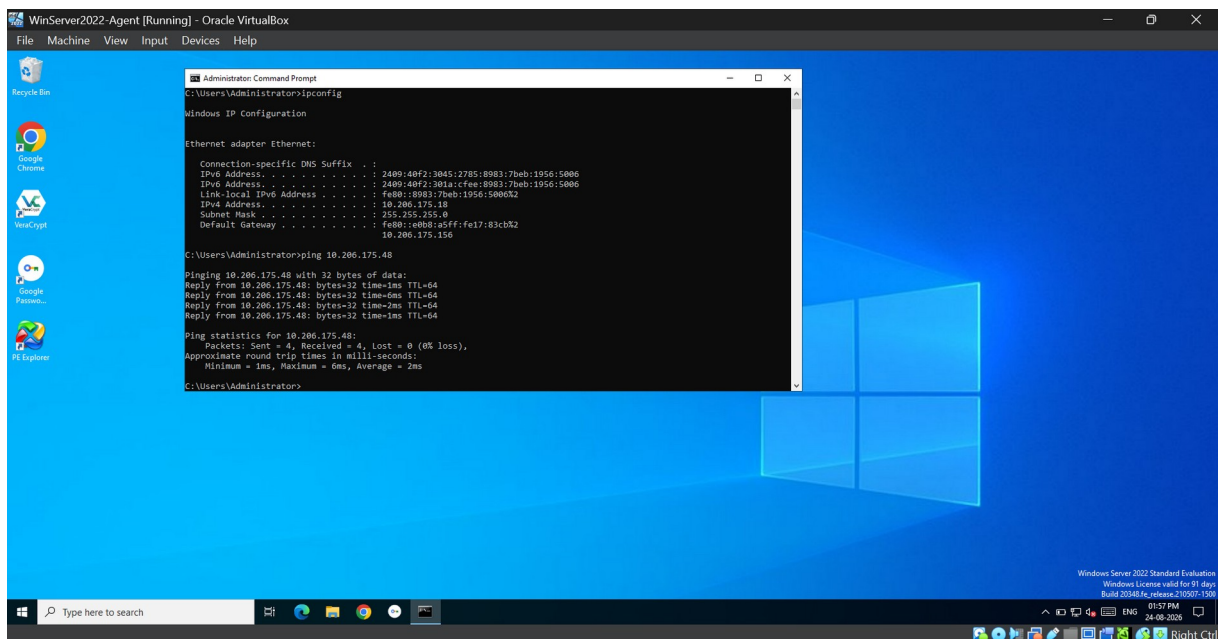
manuara@kali:~$ ping 10.206.175.18
PING 10.206.175.18 (10.206.175.18) 56(84) bytes of data:
 64 bytes from 10.206.175.18: icmp_seq=1 ttl=128 time=4.67 ms
 64 bytes from 10.206.175.18: icmp_seq=2 ttl=128 time=3.95 ms
 64 bytes from 10.206.175.18: icmp_seq=3 ttl=128 time=3.28 ms
 64 bytes from 10.206.175.18: icmp_seq=4 ttl=128 time=3.59 ms
 64 bytes from 10.206.175.18: icmp_seq=5 ttl=128 time=2.54 ms
 64 bytes from 10.206.175.18: icmp_seq=6 ttl=128 time=1.91 ms
^C
--- 10.206.175.18 ping statistics ---
 6 packets transmitted, 6 received, 0% packet loss, time 5062ms
 rtt min/avg/max/mdev = 1.907/3.292/4.469/0.843 ms

manuara@kali:~$

```

Figure 1 – Kali Linux VM IP Configuration and Connectivity Test

Figure 1 shows the output of the ip a command confirming the IPv4 address of the Kali Linux VM, along with a successful ping test to the Windows Server 2022 VM with 0% packet loss.



```

C:\Users\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : 
   IPv6 Address. . . . . : 2409:40f2:3045:2785:8983:7beb:1956:5006
   IPv6 Address. . . . . : 2409:40f2:3045:cfce:8982:7beb:1956:5006
   Link-local IPv6 Address . . . . . : fe80:8983:7beb:1956:5006%2
   IPv4 Address. . . . . : 10.206.175.48
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : fe80:a008:a5ff:fe17:83cb%2
                               10.206.175.156

C:\Users\Administrator>ping 10.206.175.48

Pinging 10.206.175.48 with 32 bytes of data:
Reply from 10.206.175.48: bytes=32 time=1ms TTL=64
Reply from 10.206.175.48: bytes=32 time=6ms TTL=64
Reply from 10.206.175.48: bytes=32 time=2ms TTL=64
Reply from 10.206.175.48: bytes=32 time=1ms TTL=64

Ping statistics for 10.206.175.48:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 6ms, Average = 2ms

C:\Users\Administrator>

```

Figure 2 – Windows Server 2022 VM IP Configuration and Connectivity Test

Figure 2 shows the output of the ipconfig command confirming the IPv4 address and subnet mask of the Windows Server 2022 VM, along with a successful ping test to the Kali Linux VM showing 0% packet loss.

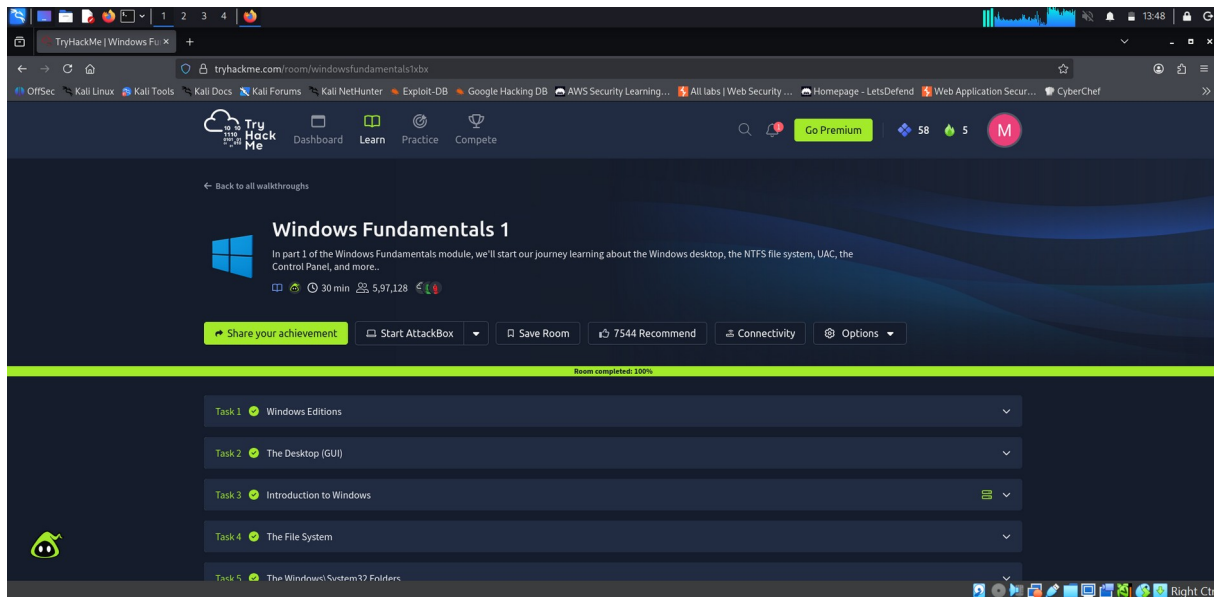


Figure 3 – TryHackMe Windows Fundamentals 1 – 100% Completion

Figure 3 shows the Windows Fundamentals 1 room with a room completion status of 100%, confirming that all required content in the room was completed.

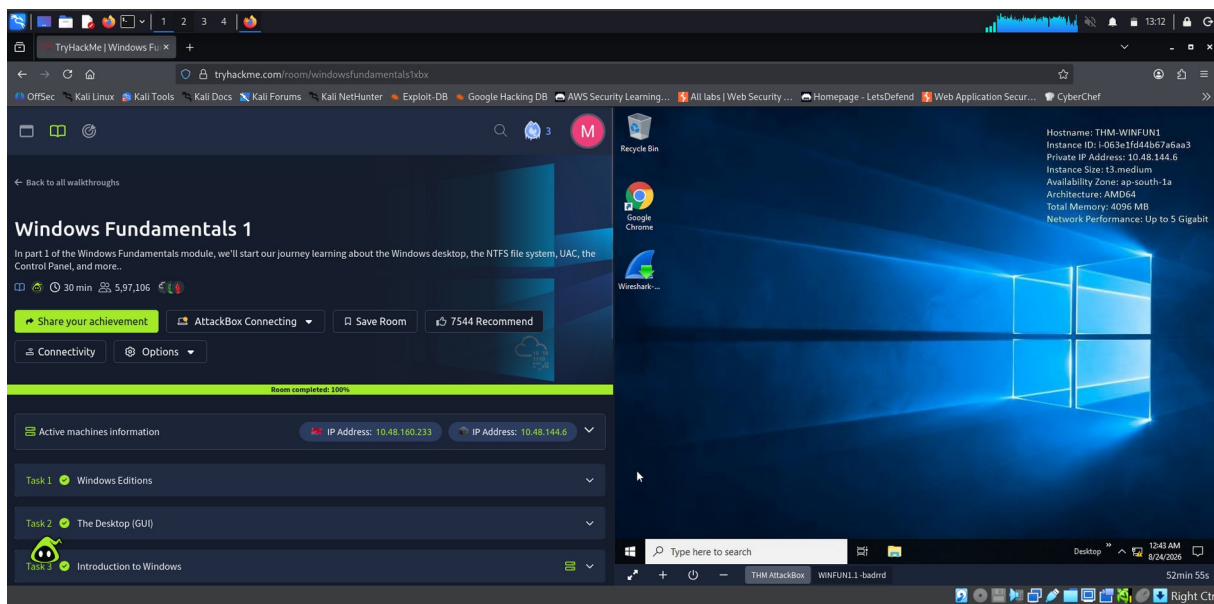


Figure 4 – TryHackMe Windows Fundamentals 1 – Completed Tasks

Figure 4 shows the individual tasks within the Windows Fundamentals 1 room, each marked with a green completion indicator, confirming that every task in the room was completed.

14. Conclusion

The existing virtual lab environment was successfully validated during this task. Bidirectional connectivity between the Kali Linux and Windows Server 2022 virtual machines was confirmed with 0% packet loss in both directions, and the required TryHackMe Windows Fundamentals 1 room was completed in full. This validated environment provides a practical foundation for upcoming SOC tasks involving monitoring, logging, analysis, and investigation.